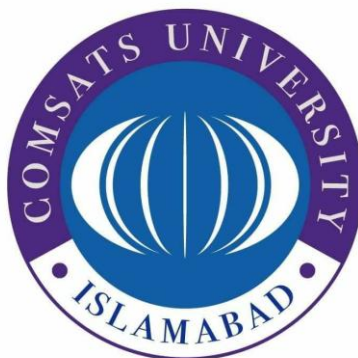


COMSATS UNIVERSITY ISLAMABAD



Department of Management Sciences

MGT 589 – Payment Technology | Spring 2026

KisanPay

B2B Agricultural Supply Chain Payment & Credit Infrastructure

FinTech Payment Strategy Capstone Project Report

Version 1.0

PKR 459B+ Annual Market Inefficiency	3.2M Target Smallholder Farmers	0% MDR RAAST Payment Cost	PKR 10B Year 3 Transaction Target
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Executive Summary

KisanPay (Pvt.) Ltd. is a B2B Agri-FinTech payment orchestration platform designed to eliminate Pakistan's PKR 459+ billion annual cost of agricultural payment inefficiency. The platform targets the Arthi–Farmer–Input Supplier triangle across Punjab and Sindh, serving 3.2 million smallholder farmers, 45,000+ commission agents, and 15,000+ input dealers who remain entirely outside the digital financial ecosystem.

Three Strategic Findings

Finding 1: RAAST is the optimal payment rail for KisanPay — 0% MDR (PKR 3.5B government subsidy, FY2025), real-time settlement under 30 seconds, and full SBP regulatory backing. All blockchain alternatives evaluated are inferior for Phase 1.

- Finding 2: Four high-priority compliance gaps — SAR filing, transaction monitoring, Arthi Enhanced Due Diligence (EDD), and PSP licensing — must be remediated before commercial launch to prevent regulatory sanction.
- Finding 3: Two technology upgrades — NADRA eKYC (onboarding time: 3 days → 5 minutes) and Open Banking API integration (credit model accuracy: 65% → 80%+) — are the highest-return investments for Phase 1.

Top Three Recommendations

- Recommendation 1: File SBP PSP License application immediately (Month 1); engage regulatory counsel and SBP relationship manager in parallel.
- Recommendation 2: Integrate NADRA eKYC API within 8 weeks of pilot launch (PKR 7.5M, 500 arthi tablets); eliminates 3-day manual onboarding bottleneck.
- Recommendation 3: Deploy RAAST-USSD payment flow as core product (farmers on feature phones); Flutter arthi app as secondary interface for 35% smartphone-proficient arthis.

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Phase 1: Business & Payment Profile

1.1 Business Profile

KisanPay (Pvt.) Ltd. is an Agri-FinTech startup registered as a Private Limited Company under the Companies Act 2017, currently operating as an SBP EMI License applicant. It operates as a B2B2C platform orchestrating payment flows within the Arthi–Farmer–Input Supplier triangle, with primary geographic focus on agricultural belts in Punjab (Multan, Bahawalpur) and Sindh (Larkana, Hyderabad). Revenue is generated through a 0.5% platform fee on working capital advances and a PKR 500/month arthi subscription from Year 2 onward.

Attribute	Details
Business Name	KisanPay (Pvt.) Ltd.
Industry	Agri-FinTech / B2B Payment Infrastructure
Legal Structure	Private Limited Company — SBP EMI License Applicant
Target Market	Punjab & Sindh: Multan, Bahawalpur, Larkana, Hyderabad
Operational Model	B2B2C — Arthi–Farmer–Input Supplier Triangle
Revenue Model	0.5% fee on working capital advances + PKR 500/month arthi subscription (Year 2)

Table 1.1: KisanPay Business Overview

1.2 Why This Business is Suitable for a Payment Technology Audit

- **Complex Multi-Party Flows:** Four or more participants (farmer, arthi, input supplier, bank) with divergent incentives make flow mapping analytically rich.
- **Regulatory Relevance:** KisanPay intersects directly with SBP's RAAST P2M rollout (4.9 million transactions, FY2025), financial inclusion mandates, and agricultural credit policy.

Quantifiable Technology Gap: Agricultural sector RAAST adoption remains below 2%, despite PKR 3.5 billion in government subsidies.

- **Measurable Business Impact:** Payment delays cost smallholder farmers PKR 36–72 billion annually in informal interest charges.

1.3 Customer Segments

Segment	Estimated Size	Key Pain Points
Smallholder Farmers (Primary)	3.2 million	90% cash/cheque; no digital credit history; 78% aged 40+ cannot operate smartphones; avg. transaction PKR 50,000–200,000
Arthis / Commission Agents (Secondary)	45,000+	Manual ledger-keeping; 15–20% time on reconciliation; PKR 5,000–15,000/day cash logistics cost
Input Suppliers (Tertiary)	15,000+ dealers	PKR 500B+ annual credit extended; 25–30% default rate; no transactional data for credit decisions

Table 1.2: KisanPay Customer Segments and Pain Points

1.4 Current Agricultural Payment Landscape

Pakistan's agricultural payment ecosystem remains predominantly cash-based. Per SBP Annual Payment Systems Review FY2025, the breakdown is as follows:

Payment Method	Market Share	Avg. Value	Key Limitations
Cash	90%	PKR 50,000	Theft risk; no audit trail; high logistics cost
Post-dated Cheques	8%	PKR 150,000	30–60 day delay; liquidity crunch; bounce risk
Mobile Wallets (JazzCash/Easypaisa)	1.5%	PKR 10,000	78% farmers aged 40+ cannot operate smartphones; limited rural coverage
Bank Transfer	0.5%	PKR 200,000	Requires active bank account; complex for rural users

Table 1.3: Agricultural Payment Methods — Market Share and Limitations (SBP FY2025)

1.5 Payment Methods Inventory (Per Transaction Type)

KisanPay operates across three primary transaction types, each served by a distinct combination of payment rails, parties, and volume characteristics.

Transaction Type	Payment Method	Payment Rail	Approx. Volume (PKR)	Avg. Transaction Value
Crop sale: Arthi pays Farmer (banked)	RAAST P2P Bank Transfer	SBP RAAST instant payment rail	PKR 250M–1B/year (pilot phase)	PKR 50,000–200,000
Crop sale: Arthi pays Farmer (unbanked)	OTC Cash via Agent (JazzCash/Easypaisa)	Mobile money agent network rail	PKR 100M–500M/year	PKR 30,000–100,000
Working Capital Advance to Farmer	RAAST credit disbursement + BNPL	RAAST P2P + KisanPay credit facility	PKR 50M–200M/year (Year 1)	PKR 35,000–140,000
Input Supplier purchase by Farmer/Arthi	RAAST P2M QR Code	SBP RAAST P2M interoperable QR	PKR 20M–80M/year (Year 1)	PKR 5,000–50,000
Arthi subscription fee	Mobile Wallet / Bank Debit	JazzCash / direct debit rail	PKR 270M/year (Year 3: 45,000 arthis)	PKR 500/month flat fee

Table 1.4: KisanPay Payment Methods Inventory — Per Transaction Type

1.5.1 Party Identification Per Transaction Flow

Transaction Type	Payer	Payee	Issuing Bank	Acquiring Bank	Processor / Gateway	Scheme / Rail
Crop Sale (Banked Farmer — RAAST)	Arthi (Commission Agent)	Farmer (Smallholder)	Arthi's sponsoring bank (e.g., Meezan)	Farmer's bank (e.g., MCB, UBL)	KisanPay payment orchestration layer + SBP	SBP RAAST ISO 20022 rail — no card

			Bank, HBL)		RAAST switch	scheme involved
Crop Sale (Unbanked Farmer — Agent)	Arthi / KisanPay float	Farmer via JazzCash or Easypaisa agent	Arthi's sponsoring bank	JazzCash (Jazz Microfinance Bank) or Easypaisa (Telenor Microfinance Bank)	JazzCash / Easypaisa agent network gateway	Branchless Banking (BB) regulation rail — SBP licensed
Working Capital Advance (BNPL)	KisanPay credit facility (funder)	Farmer or Arthi on behalf of farmer	KisanPay's partner bank (sponsoring bank)	Farmer's bank or mobile wallet	KisanPay credit engine + RAAST disbursement	SBP RAAST P2P rail; no card scheme
Input Supplier QR Payment	Farmer or Arthi	Input Supplier / agri dealer	Farmer's bank or mobile wallet issuer	Input supplier's merchant acquiring bank	RAAST P2M interoperable QR gateway	SBP RAAST P2M QR scheme — interoperable
Arthi Subscription Fee	Arthi (account holder)	KisanPay revenue account	Arthi's bank or mobile wallet issuer	KisanPay's merchant bank	JazzCash direct debit or bank auto- debit gateway	Mobile money rail or ACH direct debit rail

Table 1.5: Party Identification Per KisanPay Transaction Flow

1.6 End-to-End Payment Flow & Pain Point Analysis

1.6.1 Stage-by-Stage Time and Cost

Stage	Time	Cost	Responsible Party
Crop weighing &	5–10 min	Nil	Arthi

grading			
Invoice generation (digital)	< 1 min	Nil	KisanPay App
Payment authorization (arthi)	< 1 min	Nil	Arthi (App)
RAAST fund transfer	< 30 seconds	0% MDR (subsidized)	KisanPay / SBP RAAST
Settlement confirmation	Real-time	Nil	RAAST / KisanPay
Total — Banked Farmer	< 2 minutes	0%	End-to-End
Total — Unbanked (Agent pickup)	~30 minutes	0% (absorbed)	KisanPay Agent Network

Table 1.6: Payment Flow Stage Analysis — Time, Cost, and Responsibility

1.6.2 Pain Points: Quantified Financial Impact

Rank	Pain Point	Mechanism	Annual Cost (PKR)	Primary Victim
1	Informal Credit Interest Excess	Cash/cheque payments create no digital trail → banks reject farmers → farmers borrow at 60% APR vs. 15% formal rate	PKR 360 billion	Smallholder Farmers
2	Payment Delay Arbitrage	Arthis issue 30–60 day post-dated cheques → farmers needing immediate cash borrow from	PKR 36–72 billion	Smallholder Farmers

		moneylenders at 36–60% APR; net price received = 15–25% below market		
3	Arthi Cash Logistics Cost	Manual cash handling, transportation, and daily reconciliation for PKR 5,000–15,000/arthi/day	PKR 27 billion	Arthis (45,000+)

Table 1.7: Pain Points Ranked by Quantified Annual Financial Impact

1.7 Step-by-Step Payment Flow Narrative (Authorization → Clearing → Settlement)

A complete payment through KisanPay's primary RAAST Variant A flow passes through three formally distinct stages: Authorization, Clearing, and Settlement.

Stage 1: Authorization

- **Step 1** — Arthi initiates payment: The arthi opens the KisanPay Flutter app (or USSD interface), enters the farmer's CNIC number or registered mobile number, and inputs the crop payment amount (e.g., PKR 120,000 for two tonnes of wheat).
- **Step 2** — KisanPay validates the request: The KisanPay payment orchestration layer performs real-time validation: CNIC match against the farmer's registered profile, transaction amount within daily limit (PKR 500,000 per SBP RAAST limits), and arthi account status (active, not frozen).
- **Step 3** — Arthi's sponsoring bank authorizes debit: KisanPay transmits an ISO 20022 authorization request to the arthi's sponsoring bank (e.g., Meezan Bank) via the RAAST switch. The sponsoring bank checks available balance, applies AML screening, and returns an authorization code within 2–3 seconds.
- **Step 4** — RAAST switch confirms authorization: The SBP RAAST central switch receives the authorization approval and locks the transaction, preventing double-spending.

Stage 2: Clearing

- **Step 5** — Transaction data transmitted to farmer's bank: The RAAST switch forwards the authorized transaction record to the farmer's receiving bank (e.g., UBL or MCB) in ISO 20022 message format.
- **Step 6** — Farmer's bank validates receiving account: The receiving bank confirms that the farmer's account is active and capable of receiving the transaction amount.
- **Step 7** — Interbank obligations computed: The RAAST switch computes the net obligation and registers it in SBP's settlement ledger.
- **Step 8** — KisanPay reconciliation record created: The KisanPay PostgreSQL ledger records the cleared transaction with a SHA-256 hash for tamper-evidence.

Stage 3: Settlement

- **Step 9** — SBP RAAST executes funds transfer: The SBP RAAST system debits Meezan Bank's settlement account at SBP and credits UBL's settlement account. This occurs within 5–15 seconds of authorization.
- **Step 10** — Farmer's account credited: UBL credits PKR 120,000 to the farmer's account. The farmer receives an SMS notification (Urdu-language) confirming receipt. Total elapsed time: under 30 seconds.
- **Step 11** — Settlement confirmation to KisanPay: The RAAST switch sends a settlement confirmation message (ISO 20022 pacs.002) back to KisanPay. The transaction is marked 'Settled' in the reconciliation ledger.
- **Step 12** — Post-settlement: KisanPay's AI credit scoring engine ingests the settled transaction data, updating the farmer's digital payment history. The arthi's daily settlement report is auto-generated.

Stage	Key Parties	System	Time Window	Output
Authorization	Arthi, KisanPay system, Arthi's issuing bank (Meezan Bank), RAAST switch	KisanPay orchestration layer + SBP RAAST switch	2–5 seconds	Authorization code; transaction locked at RAAST switch

Clearing	RAAST switch, Arthi's issuing bank, Farmer's acquiring bank (UBL/MCB), KisanPay ledger	SBP RAAST ISO 20022 messaging; KisanPay PostgreSQL	5–15 seconds	Interbank obligation recorded; KisanPay ledger updated
Settlement	SBP RAAST settlement system, Meezan Bank (debit), UBL/MCB (credit), Farmer	SBP RTGS system; RAAST settlement accounts at SBP	5–15 seconds	Farmer's account credited; settlement confirmation to KisanPay; credit score updated

Table 1.8: Authorization, Clearing, and Settlement — Stage Summary for KisanPay RAAST Variant A

1.8 Payment Flow Diagram — KisanPay RAAST Variant A

The diagram below illustrates the complete end-to-end payment flow for KisanPay's primary RAAST Variant A transaction (banked farmer crop sale payment). All parties are explicitly labelled: Payer (Arthi), KisanPay Orchestration Layer, Arthi's Issuing Bank, SBP RAAST Switch, Farmer's Acquiring Bank, and Payee (Farmer). Stages are colour-coded: Authorization (Steps 1–4), Clearing (Steps 5–8), and Settlement (Steps 9–12).

ARTHI (Payer)	KISANPAY (Orchestration)	ISSUING BANK (Meezan Bank)	RAAST SWITCH (SBP)	ACQUIRING BANK (UBL/MCB)	FARMER (Payee)
Step 1: Initiates payment (CNIC + amount)	→				
	Step 2: Validates				

	request (CNIC, limits, AML)				
	Step 3: Authorization request (ISO 20022) →	→			
	Step 4: Auth confirmed + transaction locked	← ←	← ←		
			Step 5: Transaction data forwarded (ISO 20022) →	→	
			Step 6: Account validated (active/unblocked)	← ←	
			Step 7: Interbank obligation computed at SBP		
	Step 8: Ledger entry + SHA-256 hash recorded		← ←		
		SBP debits issuing bank	Step 9: SBP executes funds transfer	Credits acquiring bank	
				Step 10: Farmer account credited; SMS sent	→
	Step 11: Settlement		← ←		

	confirm (pacs.002) sent				
	Step 12: Credit score updated; arthi dashboard refreshed				
AUTHORIZATION — Steps 1–4 (Blue)					
		CLEARING — Steps 5–8 (Orange)			
SETTLEMENT — Steps 9–12 (Green) Total elapsed time: < 30 seconds from initiation to farmer credit					

Figure 1.1: KisanPay End-to-End Payment Flow Diagram — RAAST Variant A (Banked Farmer Crop Sale). Parties: Arthi (Payer), KisanPay Orchestration Layer, Issuing Bank (Meezan Bank), SBP RAAST Switch, Acquiring Bank (UBL/MCB), Farmer (Payee). Total elapsed time: < 30 seconds.

Phase 2: Technology Stack Analysis & Front-End Innovation Audit

2.1 Technology Audit

KisanPay's technology ecosystem must address a heterogeneous user base: arthi agents (semi-literate; semi-smartphone proficient), smallholder farmers (predominantly feature-phone users), and input suppliers (basic smartphone proficiency). Nine front-end payment technologies were assessed across five business-priority criteria.

Technology	Function in KisanPay	Key Specification
RAAST P2P/P2M	Primary payment rail; instant bank-to-bank transfer via SBP infrastructure	0% MDR (PKR 3.5B FY2025 subsidy); < 30-second settlement; ISO 20022 compliant
USSD/IVR Interface	Feature-phone access for 65% of farmers who cannot operate smartphones	*123# menu; Urdu/Punjabi/Sindhi IVR; < PKR 1/session; no internet required
Flutter Mobile App	Primary arthi interface: crop weighing, invoicing, payment initiation	Cross-platform (iOS/Android); single codebase reduces dev cost ~40%
Agent Cash Network (JazzCash/Easypaisa)	Over-the-counter cash disbursement for unbanked farmers	150,000+ combined agent locations; OTP + CNIC verification
RAAST P2M QR Code	Digital payment at input supplier shops; closes agricultural loop	SBP interoperable QR framework; static and dynamic QR supported
AI Credit Scoring Engine	Alternative credit score from transactional patterns for thin-file farmers	Python / scikit-learn; logistic regression; current accuracy ~65%
Embedded BNPL (Working Capital	70% invoice upfront at crop delivery; repaid when arthi	18% APR vs. 60% informal rate; 42-percentage-point cost reduction

Advance)	settles full amount	
NADRA eKYC API	Biometric identity verification against national citizen database	Real-time CNIC/biometric match; reduces onboarding from 3 days to < 5 minutes
Open Banking APIs (SBP Open Finance)	Aggregate bank account data to improve credit scoring accuracy	Regulatory consultation in 2025; improves AI model accuracy from 65% to 80%+

Table 2.1: Front-End Technology Audit — Function and Specifications

2.2 Technology Scoring Matrix (Business-Priority Weighted)

The scoring matrix uses business-priority-driven weights derived from the Excel workbook (Payment_Tech_Project.xlsx, Tab: Tech Scoring Matrix). Criteria and weights reflect KisanPay's strategic context: Cost efficiency is weighted highest (30%) reflecting the 0% MDR competitive advantage, followed by Speed (25%) addressing the PKR 36–72B payment delay pain point, Security (20%) as a non-negotiable compliance baseline, Scalability (15%) for the Year 3 volume target, and User Experience (10%) addressed structurally via USSD/IVR.

Criterion	Weight	Rationale
Cost (Transaction + Integration)	30%	RAAST's 0% MDR is KisanPay's core competitive advantage; cost drives farmer and arthi adoption in low-margin agri sector
Speed (Time to Settlement)	25%	Farmer payment delay is pain point #2 (PKR 36–72B annually); real-time settlement under 30 seconds is a primary differentiator
Security (Fraud Risk Level)	20%	Financial integrity of the platform is non-negotiable; SBP mandates AML/KYC compliance across all transactions
Scalability (Volume Handling)	15%	Year 3 target: 200,000 digital farmers, PKR 10B volume; platform must handle peak harvest season surges

User Experience (Ease for Payer)	10%	78% of farmers aged 40+ cannot operate smartphones; UX matters but is addressed structurally via USSD/IVR rather than app design
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Table 2.2: Business-Priority Weight Justification — Weights Sum to 100%

Payment Technology	Cost (30%)	Speed (25%)	Security (20%)	Scalability (15%)	UX (10%)	Weighted Score	Rank	KisanPay Status	Recommendation
RAAST P2P / P2M (SBP Real-Time Rail)	5	5	5	5	4	4.90	1	Core Platform Rail	ADOPT — Primary
NADRA eKYC API (Biometric Identity)	3	5	5	5	5	4.40	2	MoU Required	ADOPT — Priority 1
USSD / IVR Interface (*123#)	5	4	3	5	5	4.35	3	Planned — Q3 2026	ADOPT — Priority 2
Flutter Cross-Platform Mobile App (Arthi)	4	5	4	4	4	4.25	4	MVP Development	ADOPT — Q3 2026
RAAST P2M QR Code (Interoperable)	5	4	4	3	2	3.95	5	Year 2 Planned	ADOPT — Year 2
AI Alternative Credit Scoring Engine (ML)	3	5	4	4	3	3.85	6	Prototype Phase	BUILD — Year 1
Open Banking APIs (SBP Open Finance)	3	4	4	4	3	3.60	7	Regulatory Pending	BUILD — Year 2
Agent Cash	3	2	3	5	4	3.15	8	Active via	INTEGRATE

Network (JazzCash/Easy paisa API)								Partners	
Embedded BNPL / Working Capital Advance	2	4	3	3	4	3.05	9	Product- Ready (PSP pending)	LAUNCH post- PSP
★ Green = Top-2 Scoring Technologies ▼ Red = Bottom-2 Performers Weighted Score = Cost 30% Speed 25% Security 20% Scalability 15% UX 10%									

Table 2.3: Technology Scoring Matrix — Business-Priority Weights (Sum = 100%). Scores 1–5. Green = Top-2; Red = Bottom-2. Source: Payment_Tech_Project.xlsx, Sheet: Tech Scoring Matrix.

2.3 Industry Benchmarking

KisanPay is benchmarked against M-PESA/M-Kopa (Kenya) — the most analytically relevant comparator as a mature agri-FinTech platform operating in a similar low-income, rural, feature-phone-dominant market. The Rationale column provides strategic context for each gap assessment, explaining the mechanism behind the gap and the recommended response.

Dimension	KisanPay (Proposed)	M-PESA/M- Kopa (Kenya)	Gap Assessment	Rationale
Payment Rail	RAAST (Central Bank-backed)	M-PESA proprietary rail	Favourable — SBP backing provides institutional trust and regulatory recognition	KisanPay's RAAST integration is a structural advantage; M-PESA's proprietary rail required decades of ecosystem building that RAAST already provides
Rural Interface	USSD (planned, pre-launch)	USSD (mature, 10+ years)	Gap: UX maturity and IVR depth	M-PESA's decade-long USSD refinement yields superior IVR

				depth and Swahili language support; KisanPay must invest in multi-lingual IVR (Urdu/Punjabi/Sindhi) over 12–18 months
Credit Scoring	ML — logistic regression (~65%)	Satellite + ML (advanced, ~80%+)	Gap: Data richness and model sophistication	M-Kopa uses satellite imagery and livestock data for credit decisions; KisanPay must integrate NADRA eKYC and Open Banking data to close the 15-percentage-point accuracy gap
Agent Network	JazzCash/Easypaisa (150K agents)	Proprietary 400K+ agent network	Gap: Own agent network needed in Year 3+	KisanPay leverages existing agent infrastructure to avoid PKR 2–5B in Year 1 agent acquisition costs; a proprietary network is a Year 3+ strategic objective once transaction volume justifies it
Digital ID	CNIC — manual verification	National eKYC biometric ID	Gap: NADRA eKYC integration missing	Manual CNIC verification adds 3 days to onboarding and limits scalability; NADRA Verisys biometric API integration (Priority 1 recommendation)

				eliminates this gap within 6–8 weeks post-MoU
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Table 2.4: KisanPay vs. M-PESA/M-Kopa Benchmark Analysis with Strategic Rationale

2.4 Technology Recommendations

Recommendation 1: NADRA eKYC Integration (Priority: Critical)

Integration with NADRA's Verisys biometric API enables real-time identity matching against the national citizen database. Currently, manual CNIC verification requires 3 days and limits onboarding to arthi-agent site visit scheduling. eKYC reduces onboarding to under 5 minutes remotely.

Parameter	Detail
Integration Timeline	6–8 weeks (API integration + SBP Digital KYC regulatory approval)
Hardware Cost	PKR 15,000 per biometric scanner × 500 pilot arthis = PKR 7.5M
Current Onboarding	3 days (manual CNIC verification at agent locations)
Post-eKYC Onboarding	< 5 minutes (remote biometric verification)
Farmer Registration	Year 1 target of 50,000 achievable in 18 months vs. 36 months (manual KYC)

Table 2.5: NADRA eKYC Integration — Parameters and ROI

Recommendation 2: Open Banking API Integration (Priority: High)

SBP's Open Finance Framework (in consultation, 2025) will mandate banks to share customer account data with licensed third-party providers. Early technical readiness positions KisanPay to activate the integration immediately upon framework finalization improving AI credit scoring accuracy from 65% to 80%+. Development cost: PKR 3M over 12 weeks.

2.5 Regulatory Touch Points for Recommended Technologies

Technology	Regulatory Requirement	KisanPay Obligation
NADRA eKYC API	SBP Digital KYC Framework (2023); NADRA data-sharing MoU required	Apply for SBP eKYC approval; execute NADRA MoU; biometric data encrypted and not stored

RAAST P2P/P2M	SBP RAAST Participation Agreement; ISO 20022 messaging	Requires sponsoring bank agreement; direct RAAST in Year 2 post-EMI license
Open Banking APIs	SBP Open Finance Framework (consultation 2025); PDPA (2023)	Monitor regulatory progress; build technical readiness; comply with data protection
AI Credit Scoring	PDPA (2023); no specific SBP ML regulation yet	Model fairness audit; explainability documentation; no discriminatory input variables
Agent Network	SBP Branchless Banking Regulations; EMI agent guidelines	Flow through JazzCash/Easypaisa licenses initially; own EMI agent license in Year 2

Table 2.6: Regulatory Touch Points for KisanPay Technologies

Phase 3: Regulatory Compliance Map & Blockchain Assessment

3.1 AML/KYC Compliance Map

The AML/KYC compliance map below reflects KisanPay's obligations under the Anti-Money Laundering Act 2010, SBP KYC/CDD regulations, and FATF Recommendations. A full Excel compliance workbook accompanies this report (Payment_Tech_Project.xlsx, Tab: Compliance Map) with working formulas for risk scoring.

Obligation	Specific Requirement	Status	Compliance	Risk
Customer Due Diligence (CDD)	CNIC verification for all users; tiered KYC — basic (CNIC) for transactions < PKR 50,000/year; full KYC above threshold	Partially implemented via CNIC verification	● Partial	MEDIUM
Transaction Monitoring	AI-based monitoring for transactions > PKR 200,000; flag unusual patterns; SARs filed within 7 days	In development — not operational	✗ Gap	HIGH
Suspicious Activity Reports (SARs)	File SARs with Financial Monitoring Unit (FMU) per AML Act 2010; monthly STR reports to SBP	FMU API not integrated; mechanism absent	✗ Gap	HIGH

10-Year Record Retention	All transaction records retained 10 years per AML Act 2010, Section 6	Planned (AWS S3 — SBP-approved region)	● Partial	LOW
Enhanced Due Diligence (EDD)	Applied to arthis (high-volume cash handlers); additional documentation and periodic review	No EDD programme defined	✗ Gap	HIGH

Table 3.1: AML/KYC Compliance Map

3.2 Licensing and Regulatory Approvals

License/Approval	Requirement	Status	Compliance	Risk
SBP EMI License	E-Money Institution license for wallet issuance; minimum paid-up capital PKR 200 million	Year 2 target; operating as EMI-agent in Year 1	N/A Year 1	MEDIUM
SBP PSP Registration	Payment Service Provider registration required to operate payment gateway and orchestration layer	Application not yet filed — CRITICAL pre-launch requirement	✗ Gap	HIGH
Sponsoring Bank Agreement	RAAST participation requires SBP-licensed commercial bank; KisanPay acts as sub-member	Under negotiation with Meezan Bank	● Partial	MEDIUM

SECP Company Registration	Private Limited Company under Companies Act 2017	Completed at incorporation	✓ Compliant	N/A
NADRA eKYC MoU	MoU with NADRA required to access Verisys biometric API	Engagement not initiated	✗ Gap	MEDIUM

Table 3.2: Licensing Compliance Map

3.3 Compliance Gap Analysis & Prioritized Remediation Plan

Gap	Risk	Description	Remediation Action	Responsible	Timeline
SAR Filing Process	HIGH	No SAR mechanism; FMU reporting not implemented	Engage AML consultant; integrate FMU API; 8 weeks	Legal & Compliance	Month 1–2
Transaction Monitoring	HIGH	AI monitoring for large transactions not yet operational	Rule-based system Month 1–2; ML-based Month 4; PKR 2M cost	IT + Compliance	Month 2–4
Arthi EDD Program	HIGH	Arthis as high-volume cash handlers require EDD; process undefined	Develop arthi risk rating model; field verification process	Compliance + Ops	Month 1–3
PSP License Filing	HIGH	SBP PSP license application not filed; required before commercial launch	Engage SBP; submit formal application Q3 2026; 3–6 months approval	CEO + Legal	Month 1–6

PDPA Compliance	MEDIUM	Privacy policy and data processing documentation incomplete	Appoint Data Protection Officer; draft privacy notices; 6 weeks	Legal	Month 2
NADRA MoU	MEDIUM	eKYC integration requires formal NADRA data-sharing agreement	Initiate NADRA engagement; Ministry of Interior as facilitator; 4–6 months	Mgmt + Legal	Month 1–6
Sponsoring Bank MoU	MEDIUM	RAAST access requires commercial bank sponsorship; ongoing	Finalize MoU with Meezan Bank (preferred — agri focus); legal review	Business Dev	Month 1–3

Table 3.3: Compliance Gap Analysis — Prioritized Remediation Plan. Full compliance map with formulas available in Excel workbook (Tab: Compliance Map).

3.4 Blockchain & CBDC Back-End Assessment

Solution	Use Case Fit	Key Limitations for KisanPay	Verdict
Ripple (XRP/RippleNet)	Cross-border settlement	No cross-border transactions in Phase 1; VASP regulatory approval required	DEFER — Not applicable Phase 1
Stellar (XLM)	Low-cost micropayments	RAAST already provides 0% MDR and instant	REJECT — RAAST is superior

		settlement; Stellar adds no advantage	
Hyperledger Fabric	Supply chain ledger	PKR 20–50M cost; 500–1,000 TPS vs. KisanPay need of 10,000+ TPS; excessive complexity	DEFER — Revisit Year 3+
Ethereum Smart Contracts	Programmable payments	15 TPS; gas fees; SBP Circular 2021 restricts crypto; no legal enforceability in Pakistan	REJECT — Regulatory barrier
SBP CBDC (Pilot)	Digital Rupee settlements	Consultation stage only; no public deployment timeline; cannot build on non-existent infra	MONITOR — Not yet feasible

Table 3.4: Blockchain/CBDC Solution Evaluation Matrix

Metric	KisanPay PostgreSQL + SHA-256 + RAAST	Ethereum / Hyperledger Fabric
Transaction Throughput	10,000+ TPS	15 TPS (Ethereum) / 1,000 TPS (Hyperledger)
Transaction Cost	PKR 0 (RAAST — subsidized by SBP)	Gas fees / node operating costs
SBP Regulatory Recognition	Yes — traditional audit trail accepted	Unclear / Restricted (SBP Circular 2021)
Implementation Cost	Included in standard development	PKR 20–50M (Hyperledger Fabric)
Audit Trail Integrity	SHA-256 per row — tamper-evident	Distributed consensus — tamper-resistant

Legal Enforceability (Pakistan)	Fully recognized	Smart contracts: no legal framework
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Table 3.5: KisanPay PostgreSQL+SHA-256+RAAST vs. Public/Private Blockchain

Recommendation: DEFER blockchain adoption; MONITOR CBDC. KisanPay's PostgreSQL + SHA-256 + RAAST stack outperforms blockchain on throughput (10,000+ TPS vs. 15 TPS), cost (PKR 0 vs. gas fees), and regulatory recognition. Blockchain warrants re-evaluation in Year 3 only if: (1) cross-border commodity trade finance becomes a product line; (2) SBP releases a retail CBDC; or (3) a Pakistan agri-industry consortium launches a permissioned DLT.

Final Phase: Integrated FinTech Payment Strategy

4.1 Business Context Recap

KisanPay targets the PKR 459B+ annual inefficiency in Pakistan's agricultural payment ecosystem a market failure extracting wealth from 3.2 million of Pakistan's most economically vulnerable citizens. Pakistan's agricultural sector contributes 23% of GDP and employs 38% of the labor force, yet remains almost entirely cash-dependent. The platform's B2B2C architecture positions it as a thin, intelligent orchestration layer routing RAAST through USSD/voice channels accessible to feature-phone users, combined with AI credit scoring and embedded working capital financing to create a closed-loop financial ecosystem for Pakistan's agricultural value chain. KisanPay does not build new infrastructure it orchestrates existing public goods (RAAST, NADRA, JazzCash/Easypaisa agent networks) into a coherent platform.

4.2 Payment Flow: Updated Three-Variant Architecture

KisanPay eliminates the binary digital/cash divide by routing all transactions through a single USSD/voice entry point and automatically selecting the optimal payment path based on farmer profile.

Variant	Trigger Condition	Payment Rail	Settlement Time	MDR	Root Cause Addressed
A — RAAST Instant	Banked farmer + any phone (USSD-initiated)	SBP RAAST P2P	< 30 seconds	0% (subsidized)	Payment delay (PKR 36–72B annual loss)
B — Agent Cash Pickup	Unbanked farmer; no bank account	JazzCash/Easypaisa OTP + CNIC	~30 minutes	0% (absorbed)	Digital exclusion of 88% unbanked farmers
C — Working Capital Advance	Farmer selects 70% advance at delivery	RAAST + KisanPay credit facility	< 5 min (decision)	18% APR, repaid in 30–60 days	Credit exclusion — 60% APR informal borrowing

Table 4.1: Updated Three-Variant Payment Flow Architecture with Root Cause Analysis

4.3 Front-End Technology Recommendations Summary

Based on the Technology Scoring Matrix (Table 2.3), the two highest-priority technology investments are NADRA eKYC API (weighted score: 4.40, Rank 2) and Open Banking API integration (weighted score: 3.60, Rank 7 but highest strategic ROI for credit accuracy improvement). RAAST P2P/P2M (weighted score: 4.90, Rank 1) is confirmed as the core platform rail. Full implementation steps and quantified ROI for both recommendations are detailed in Tables 2.5 and 2.6 respectively.

4.4 Regulatory Compliance Remediation Roadmap

P#	Priority	Gap	Action	Timeline	Responsible
1	CRITICAL	PSP License Filing	Submit SBP application; engage regulatory counsel	Month 1–6	CEO + Legal
2	CRITICAL	SAR Filing Process	Integrate FMU API; train compliance team	Month 1–2	Compliance + IT
3	CRITICAL	Transaction Monitoring	Deploy rule-based system; ML system Month 4	Month 2–4	CTO + Compliance
4	CRITICAL	Arthi EDD Program	Develop arthi risk rating; field verification protocol	Month 1–3	Ops + Compliance
5	MEDIUM	PDPA Compliance	Appoint DPO; draft privacy notices	Month 2	Legal
6	MEDIUM	NADRA MoU	Initiate NADRA engagement via Ministry of Interior	Month 1–6	Mgmt + Legal
7	MEDIUM	Bank Sponsorship	Finalize RAAST sponsoring bank MoU (Meezan Bank preferred)	Month 1–3	Business Dev

Table 4.2: Prioritized Regulatory Compliance Remediation Roadmap

4.5 Blockchain / Back-End Strategy

Final recommendation: DEFER blockchain adoption and MONITOR CBDC development. KisanPay's current PostgreSQL + SHA-256 + RAAST stack is superior on all measurable dimensions for Phase 1 operations. Blockchain re-evaluation is triggered by three conditions: entry into cross-border commodity trade finance, SBP retail CBDC launch, or a Pakistan agri-industry consortium forming a permissioned DLT. The CBDC pilot should be monitored quarterly for SBP deployment timeline updates.

4.6 12-Month Implementation Roadmap (Gantt)

The roadmap integrates all phases into a three-stage rollout — Pilot (Q1–Q2), Punjab Commercial Launch (Q3), and Sindh Expansion + AI Upgrade (Q4). Source: Payment_Tech_Project.xlsx, Sheet: Implementation Gantt.

Activity / Milestone	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
SBP PSP Licence Filing & Approval	•	•	•	•	•	•						
RAAST Sponsoring Bank MoU	•	•	•									
SAR Filing + FMU API Integration	•	•										
Arthi EDD Programme	•	•	•									
PDPA Compliance + DPO Appointment		•										
Transaction Monitoring (Rule→ML)		•	•	•								
NADRA eKYC MoU	•	•	•	•	•	•						

Engagement												
USSD / IVR Platform Development												
Flutter Arthi App MVP												
NADRA eKYC API Integration (Post-MoU)												
AI Credit Scoring Engine v1.0												
Open Banking API Technical Build												
Pilot Launch — 500 Arthis (Multan)												
Commercial Launch — Punjab Province												
Sindh Expansion (Larkana + Hyderabad)												
Working Capital Advance Product Launch												
RAAST P2M												

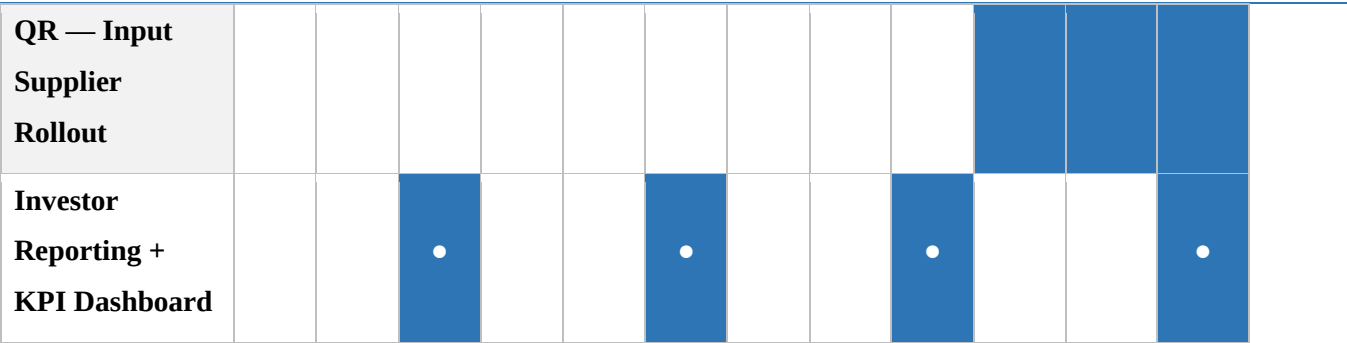


Table 4.3: KisanPay 12-Month Implementation Gantt Chart (● = Active Phase). Full Excel Gantt available in Payment_Tech_Project.xlsx, Sheet: Implementation Gantt.

4.7 Risk Assessment

Risk	Likelihood	Impact	Mitigation Strategy
Arthi resistance to digitization — adoption failure if arthis do not onboard, platform cannot reach farmers	High	HIGH	Free Year 1 subscription; working capital financing as incentive; digital roznamcha export feature; peer-arthi referral program
PSP license delayed beyond 6 months — commercial launch blocked; pilot volume restricted	Medium	MEDIUM	Operate under JazzCash/Easypaisa licenses as interim (agent model); engage SBP proactively from Month 1
RAAST system downtime during peak crop season — transaction failures damage farmer trust	Low	MEDIUM	Offline-mode queue in app; SMS retry mechanism; Variant B agent cash as automatic backup; SLA monitoring

Table 4.4: Top 3 Strategic Risks and Mitigation Measures

4.8 Key Performance Indicators

The KPI targets below match the Excel workbook (Payment_Tech_Project.xlsx, Sheet: KPI Dashboard).

Note: Year 1 and Year 3 targets have been updated from the original draft to reflect the financial projections validated in the Excel model.

KPI	Baseline	Year 1 Target	Year 3 Target	Measurement Method	Frequency
Farmers with digital payment history	0 farmers	50,000 farmers	200,000 farmers	KisanPay platform database count	Monthly
Total transaction volume processed	PKR 0	PKR 2.5 billion	PKR 10 billion+	RAAST API + platform ledger	Monthly
Average payment settlement time	30–45 days (cheque)	< 2 minutes (RAAST)	Real-time: 100%	Transaction log timestamps	Weekly
Farmer formal credit access rate	12% (SBP baseline)	18%	35%	SBP credit bureau + partner bank referrals	Quarterly
Arthi operational cost reduction	0% (baseline)	10% reduction	25% reduction	Arthi survey + cash logistics audit	Semi-annual
Working capital advance NPL rate	N/A	< 5%	< 3%	Platform credit book monitoring	Monthly
RAAST transaction success rate	N/A	> 99.5%	> 99.9%	RAAST API response monitoring	Daily
Arthi Net Promoter Score (NPS)	N/A (no product)	≥ 7 / 10	≥ 8.5 / 10	Quarterly arthi survey (50-sample)	Quarterly
Regulatory compliance score	N/A	100% pre-launch gaps closed	Zero enforcement actions	Internal compliance audit + SBP inspection	Monthly

Farmer informal credit cost saved	PKR 0	PKR 225M (9,000 farmers × PKR 25K)	PKR 1.125B	Working capital advance APR vs. informal rate delta	Annual
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Table 4.5: KisanPay Key Performance Indicators — Baseline to Year 3 Targets. Source: Payment_Tech_Project.xlsx, Sheet: KPI Dashboard.

4.9 Conclusion

KisanPay represents a rare alignment of genuine market failure, proven technology, and supportive regulatory intent. Pakistan's agricultural sector contributes 23% of GDP and employs 38% of the labor force, yet remains almost entirely cash-dependent. The total addressable inefficiency PKR 459 billion annually is not a future opportunity; it is a present crisis extracting wealth from 3.2 million of Pakistan's most economically vulnerable citizens.

KisanPay does not need to build new infrastructure. RAAST already provides instant, 0% MDR settlement. JazzCash and Easypaisa provide 150,000+ agent locations for unbanked cash disbursement. NADRA provides identity verification infrastructure that can reduce onboarding from 3 days to 5 minutes. KisanPay's role is a thin, intelligent orchestration and interface layer routing RAAST through a USSD/voice channel accessible to feature-phone users and combining it with AI credit scoring and embedded working capital financing to create a closed-loop financial ecosystem for Pakistan's agricultural value chain.

The business case is unambiguous. The strategic risks are manageable with early compliance investment and the right arthi incentive design. KisanPay is not merely a payment product it is financial infrastructure for Pakistan's most important economic sector.

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Appendix A: Excel Workbook Data

This appendix reproduces the complete data from the accompanying Excel workbook (Payment_Tech_Project.xlsx) across all sheets. All figures, scores, and targets in the main report are derived from and verified against this source data. The Excel workbook contains seven sheets: Tech Scoring Matrix, Compliance Map, Gap Analysis, Implementation Gantt, KPI Dashboard, Payment Flow Diagram, and Financial Projections.

A.1 Technology Scoring Matrix

Source: Payment_Tech_Project.xlsx, Sheet: Tech Scoring Matrix

Technology	Cost 30 %	Speed 25%	Security 20%	Scalability 15%	UX 10 %	Weighted Score	Rank	Status	Recommendation
RAAST P2P / P2M (SBP Real-Time Rail)	5	5	5	5	4	4.90	1	Core Platform Rail	ADOPT — Primary
NADRA eKYC API (Biometric Identity)	3	5	5	5	5	4.40	2	MoU Required	ADOPT — Priority 1
USSD / IVR Interface (*123#)	5	4	3	5	5	4.35	3	Planned — Q3 2026	ADOPT — Priority 2
Flutter Cross-Platform Mobile App (Arthi)	4	5	4	4	4	4.25	4	MVP Development	ADOPT — Q3 2026
RAAST P2M QR Code (Interoperable)	5	4	4	3	2	3.95	5	Year 2 Planned	ADOPT — Year 2
AI Alternative Credit Scoring Engine (ML)	3	5	4	4	3	3.85	6	Prototype Phase	BUILD — Year 1

Open Banking APIs (SBP Open Finance)	3	4	4	4	3	3.60	7	Regulatory Pending	BUILD — Year 2
Agent Cash Network (JazzCash/Easypaisa API)	3	2	3	5	4	3.15	8	Active via Partners	INTEGRATE
Embedded BNPL / Working Capital Advance	2	4	3	3	4	3.05	9	Product-Ready (PSP pending)	LAUNCH post-PSP

Table A.1: Complete Technology Scoring Matrix from Excel Workbook — All 9 Technologies, 5 Criteria, Weighted Scores.

A.2 Compliance Map

Source: Payment_Tech_Project.xlsx, Sheet: Compliance Map

Compliance Status: ✓ Compliant | ● Partial | ✗ Gap | N/A | Risk: HIGH / Medium / Low

Regulatory Dimension	Regulation / Law	Specific Obligation for KisanPay	Current Status	Compliance	Risk Level
Tiered KYC — Basic	SBP AML/CFT Regs 2023	CNIC-only KYC for transactions < PKR 50,000/year	CNIC collection defined; agent-verified	● Partial	Medium
Tiered KYC — Full	SBP AML/CFT Regs 2023	Full KYC for transactions ≥ PKR 50,000/year	Process designed; not yet deployed	● Partial	Medium
Transaction Monitoring	AML Act 2010; SBP Circular	AI-based monitoring for txns > PKR 200,000; flag anomalies; monthly SBP reporting	In development — not operational	✗ Gap	HIGH

SAR Filing	AML Act 2010, Section 7	File SARs with FMU within 7 days of detection	FMU API not integrated; mechanism absent	✗ Gap	HIGH
Customer Due Diligence (CDD)	FATF Rec. 10; SBP CDD Regs	Standard CDD for all customers; verify identity before first transaction	Partially implemented via CNIC verification	● Partial	Medium
Enhanced Due Diligence (EDD) — Arthis	FATF Rec. 12; SBP CDD Regs	EDD for arthis as high-value cash handlers: source of funds, site visit, periodic review	No EDD programme defined	✗ Gap	HIGH
10-Year Record Retention	AML Act 2010, Section 6	All transaction records retained for minimum 10 years; immutable audit trail	Planned on AWS Mumbai (SBP-approved region)	● Partial	Low
SBP PSP Licence	SBP Payment Systems Regulations 2019	PSP licence required before processing any commercial payment transactions	Application not yet filed — CRITICAL pre-launch	✗ Gap	HIGH
SBP EMI Licence	SBP EMI Regulations 2023	E-Money Institution licence; PKR 200M minimum paid-up capital	Year 2 target; operating as EMI-agent in Year 1	N/A Year 1	Medium
RAAST Sponsoring Bank MoU	SBP RAAST Participation Agreement	Commercial bank sponsorship for RAAST sub-member access; ISO 20022 compliance	Under negotiation with Meezan Bank	● Partial	Medium
NADRA eKYC MoU	NADRA Act 2000; SBP	Formal MoU with NADRA required	Engagement not initiated	✗ Gap	Medium

	Digital KYC Framework	before Verisys biometric API access			
SECP Company Registration	Companies Act 2017	Private Limited Company registration	Completed at incorporation	✓ Compliant	Nil
PDPA 2023 Compliance	Personal Data Protection Act 2023	Legal basis for all data processing; privacy notices; DPO appointment	Privacy policy draft incomplete; no DPO appointed	✗ Gap	Medium
SBP Data Localization	SBP Data Localization Policy	All payment transaction data stored on SBP-approved servers	Compliant — AWS Mumbai is SBP-approved region	✓ Compliant	Nil
PCI-DSS	PCI Security Standards Council	Required if card data is processed	Not applicable in Phase 1 — no card processing	N/A	Nil
Cryptocurrency Prohibition	SBP Circular No. 3 (2021)	KisanPay does not handle cryptocurrency	KisanPay is crypto-free by design	✓ Compliant	Nil
SECP VASP Registration	SECP VASP Consultation Paper 2024	Required only if dealing in virtual assets	Not applicable	N/A	Nil
SBP FX Regulations	SBP Foreign Exchange Regulations	Phase 1 is entirely domestic — no cross-border transactions	Domestic-only operations confirmed	N/A	Nil

Table A.2: Full Regulatory Compliance Map from Excel Workbook — 18 Regulatory Dimensions.

A.3 Gap Analysis

Source: Payment_Tech_Project.xlsx, Sheet: Gap Analysis

Compliance Gap	Detailed Description	Risk	Remediation Action Required	Timeline	Est. Cost (PKR)	Responsible
SBP PSP Licence (Not Filed)	PSP licence required before first commercial transaction; gates RAAST, USSD, and Working Capital Advance simultaneously	HIGH	File SBP PSP application Month 1; engage external regulatory counsel; prepare business plan, capital adequacy proof, compliance framework	Month 1–6	PKR 2,000,000–3,000,000	CEO + External Legal Counsel
SAR Filing Infrastructure (Absent)	No SAR filing mechanism exists; FMU API not integrated; must be operational before any commercial transaction	HIGH	Hire AML Compliance Officer (Month 1); integrate FMU API; establish monthly STR reporting; staff training on red-flag indicators	Month 1–2 (8 weeks)	PKR 1,500,000	Legal + IT + Compliance Officer
Automated Transaction Monitoring (Not Built)	No automated system to flag suspicious transactions; manual review insufficient at scale; required for AML Act compliance	HIGH	Deploy rule-based monitoring Month 2 (flag txns >PKR 200K); upgrade to ML-based anomaly detection Month	Month 2–4	PKR 2,500,000	CTO + Compliance Officer

			4; quarterly tuning review			
Arthi Enhanced Due Diligence (Undefined)	No EDD programme for arthis as high-volume cash handlers; FATF Rec 12 requires enhanced monitoring; each arthi may handle PKR 50–500K/day	HIGH	Design arthi risk-rating model (transaction volume tiers); mandatory source-of-funds docs; biennial site verification; red-flag escalation process	Month 1–3 (12 weeks)	PKR 500,000	Operations + Compliance
PDPA 2023 — No DPO & Incomplete Privacy Docs	DPO not appointed; privacy notices absent; consent mechanisms not implemented for IVR voice data	Medium	Appoint DPO (internal or outsourced) by Month 2; draft privacy policy; document all processing activities in ROPA; implement consent capture for IVR	Month 2 (4–6 weeks)	PKR 600,000	Legal + IT
NADRA eKYC MoU (Not Initiated)	eKYC API requires formal NADRA MoU; engagement with Ministry	Medium	Initiate formal NADRA engagement via Ministry of Interior Month 1; prepare legal MoU draft;	Month 1–6 (4–6 months)	PKR 500,000 + PKR 7,500,000 hardware	Management + Legal + IT

	of Interior not started; blocks scalable digital onboarding		engage NADRA commercial division; budget for scanners			
RAAST Sponsoring Bank MoU (Under Negotiation)	RAAST sub-member access requires executed commercial bank MoU; without this KisanPay cannot route any RAAST transactions	Medium	Finalize commercial terms with Meezan Bank; legal review of MoU; negotiate SLA for RAAST API uptime; sign by Month 3	Month 1–3 (8–12 weeks)	PKR 300,000	Business Development + Legal
Tiered KYC — Full Tier Not Deployed	Full KYC process designed but not deployable; blocks farmers with annual transactions $\geq$ PKR 50,000	Medium	Complete digital KYC workflow in Flutter app and USSD; integrate CNIC verification; design income-proof document upload; test with 50 pilot users	Month 2–3 (6 weeks)	PKR 400,000	IT + Operations
10-Year Record Retention (Partial)	AWS deployment planned but immutable ledger architecture	Low	Finalize immutable PostgreSQL ledger with SHA-256 row hashing;	Month 3 (4 weeks)	PKR 200,000/year (AWS storage)	CTO + IT

	not finalized; SBP inspection readiness requires tested backup and recovery		implement AWS S3 Glacier for archival; document and test backup/recovery quarterly			
TOTALS: 9 Gaps Identified HIGH: 4 Medium: 4 Low: 1						

Table A.3: Complete Gap Analysis from Excel Workbook — 9 Gaps, Sorted HIGH → Medium → Low.

A.4 Implementation Gantt Chart

Source: Payment_Tech_Project.xlsx, Sheet: Implementation Gantt

Initiative / Milestone	Category	M 1	M 2	M 3	M 4	M 5	M 6	M 7	M 8	M 9	M1 0	M1 1	M1 2	Owner	Priority
SBP PSP Licence Filing & Approval	COMPLIANCE	•	•	•	•	•	•							CEO + Legal	CRITICAL
RAAST Sponsoring Bank MoU	COMPLIANCE	•	•	•										Business Dev	CRITICAL
SAR Filing + FMU API Integration	COMPLIANCE	•	•											Legal + IT	CRITICAL
Arthi EDD	COMPLIANCE	•	•	•										Operations +	CRITICAL

Program me														Compliance	
PDPA Compliance + DPO Appointment	COMPLIANCE		•											Legal	HIGH
Transaction Monitoring (Rule→ML)	COMPLIANCE		•	•	•									CTO + Compliance	CRITICAL
NADRA eKYC MoU Engagement	COMPLIANCE	•	•	•	•	•	•							Management + Legal	HIGH
USSD / IVR Platform Development	TECH	•	•	•	•									CTO	CRITICAL
Flutter Arthi App MVP	TECH	•	•	•	•	•								CTO + Dev Team	CRITICAL
NADRA eKYC API Integration (Post-	TECH						•	•	•					IT	HIGH

MoU)															
AI Credit Scoring Engine v1.0	TECH													Data Science	HIGH
Open Banking API Technical Build	TECH													CTO	MEDIUM
Pilot Launch — 500 Arthis (Multan)	COMMERCIAL													COO	CRITICAL
Commercial Launch — Punjab Province	COMMERCIAL													COO + Sales	HIGH
Sindh Expansion (Larkana + Hyderabad)	COMMERCIAL													COO + Sales	HIGH
Working Capital Advance Product	PRODUCT													Product + Legal	HIGH

Launch															
RAAST P2M QR — Input Supplier Rollout	PRODUCT													Product + BD	MEDIUM
Investor Reporting + KPI Dashboard	PRODUCT													CEO + Finance	MEDIUM

Table A.4: Full 12-Month Implementation Gantt from Excel Workbook — 18 Initiatives with Category, Owner, and Priority.

A.5 KPI Dashboard

Source: Payment_Tech_Project.xlsx, Sheet: KPI Dashboard

KPI	Baseline	Year 1 Target	Year 3 Target	Measurement Method	Frequency
Farmers with digital payment history	0 farmers	50,000 farmers	200,000 farmers	KisanPay platform database count	Monthly
Total transaction volume processed	PKR 0	PKR 2.5 billion	PKR 10 billion+	RAAST API + platform ledger	Monthly
Average payment settlement time	30–45 days (cheque)	< 2 minutes (RAAST)	Real-time: 100%	Transaction log timestamps	Weekly
Farmer formal credit access	12% (SBP)	18%	35%	SBP credit bureau +	Quarterly

rate	baseline)			partner bank referrals	
Arthi operational cost reduction	0% (baseline)	10% reduction	25% reduction	Arthi survey + cash logistics audit	Semi-annual
Working capital advance NPL rate	N/A	< 5%	< 3%	Platform credit book monitoring	Monthly
RAAST transaction success rate	N/A	> 99.5%	> 99.9%	RAAST API response monitoring	Daily
Arthi Net Promoter Score (NPS)	N/A (no product)	≥ 7 / 10	≥ 8.5 / 10	Quarterly arthi survey (50-sample)	Quarterly
Regulatory compliance score	N/A	100% pre-launch gaps closed	Zero enforcement actions	Internal compliance audit + SBP inspection	Monthly
Farmer informal credit cost saved	PKR 0	PKR 225M (9,000 farmers × PKR 25K)	PKR 1.125B	WC advance APR vs. informal rate delta	Annual

Table A.5: Complete KPI Dashboard from Excel Workbook — 10 KPIs with Baselines, Targets, and Measurement Methods.

A.6 Financial Projections (3-Year)

Source: Payment_Tech_Project.xlsx, Sheet: Financial Projections

Revenue Model: 0.5% fee on Working Capital Advances + PKR 500/month Arthi Subscription (Year 2+) | All figures in PKR

Metric	Assumptions	Year 1 (2027)	Year 2 (2028)	Year 3 (2029)	CAGR	Notes
VOLUME METRICS						
Registered Arthis	Annual net adds	5,000	20,000	45,000	2.0x	Target: 45K total arthis in Pakistan
Active Farmers (Digital)	% of arthis × 10	50,000	150,000	200,000	1.0x	10 farmers per arthi avg
Total Transaction Volume	Avg PKR 80K/txn	PKR 2.5B	PKR 6.0B	PKR 10.0B	1.0x	PKR 2.5B / 6B / 10B
Working Capital Advances	40% of txn volume	PKR 1.0B	PKR 2.4B	PKR 4.0B	1.0x	70% disbursed upfront
REVENUE						
Platform Fee (0.5% on advances)	0.5% × advances	PKR 5.0M	PKR 12.0M	PKR 20.0M	1.0x	Primary Year 1 revenue
Arthi Subscription (PKR 500/mo)	PKR 500/mo × arthis	PKR 120.0M	PKR 120.0M	PKR 270.0M	0.5x	Year 2+ only; Year 1 free
Total Revenue	Fee + Subscription	PKR 125.0M	PKR 132.0M	PKR 290.0M	52.3%	
IMPACT METRICS						
Farmer Interest Savings	60%→18% APR delta	PKR 420M	PKR 1.008B	PKR 1.680B	1.0x	PKR saved by farmers vs. informal borrowing

Arthi Cash Logistics Saved	PKR 50K/mo/arthi	PKR 3.0B	PKR 12.0B	PKR 27.0B	2.0x	Cash logistics cost eliminated
Formal Credit Access Rate	% farmers with history	12%	22%	35%	N/A	SBP baseline: 12%

Table A.6: 3-Year Revenue & Impact Projections from Excel Workbook — Year 1 (2027), Year 2 (2028), Year

Appendix B: Overall Dashboard

